

MCA 1st Semester Examination, 2023 (R & B)

Subject : MCA

(Operating System)

Paper —1.1.5

Full Marks : 80

Time : 3 hours

Answer all questions

The figures in the right-hand margin indicate marks

Candidates are required to give their answers in their own words as far as practicable

PART — I

1. What is OS? Briefly explain different services of OS. Briefly explain PCB. 4 + 10 + 6

Or

- (a) In FCFS : Criteria : Arrival time
Mode : Non-pre emptive

(Turn Over)

Calculate average waiting time for the following : 10

Process	A T	B T
P1	2	6
P2	5	3
P3	1	8
P4	0	3
P5	4	4

(b) Solve the SJF scheduling as per following criteria. 10

Criteria : Burst time

Mode : Pre emptive

PID	A T	B T
P1	0	8
P2	1	4
P3	2	2
P4	3	1
P5	4	3
P6	5	2

PART – II

2. What is Bankers algorithm ? Using Backer's algorithm, solve the Total Rosource A = 10, B = 5, C = 7 and show the path how to avoid deadlock. 4 + 10 + 6

Process	Allocation			Maxneed			Available	Remaining
P ₁	0	1	0	7	5	3		
P ₂	2	0	0	3	2	2		
P ₃	3	0	2	9	0	2		
P ₄	2	1	1	4	2	2		
P ₅	0	0	2	5	3	3		

Or

- (a) Differentiate between logical and physical address. 10
- (b) What is paging ? Differentiate between frame and paging along with example. 10

PART – III

3. What is demand paging ? Why swap in and swap out is required ? Using FIFO page replacement algorithm find hit and miss ratio of the following strings 20

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 1, 2, 0.

Or

- (a) What is Kernel ? Briefly explain different types of Kernel. 10
- (b) What is File allocation in OS. Explain briefly. 10

PART – IV

4. What is Swap Space Management ? Briefly explain secondary storage structure. Give example of disk scheduling. 4 + 10 + 6

Or

- (a) Briefly explain Principle of Security in OS. 10
- (b) Write the following shell code (UNIX) for 10
- (i) Copy a file
 - (ii) Current login of User
 - (iii) Change password
 - (iv) Change directory.
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